IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART I: MEng, BEng and ACGI

Corrected Copy

MATHEMATICS 1A (E-STREAM AND I-STREAM)

Thursday, 23 May 10:00 am

Time allowed: 2:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks (25% each).

NO CALCULATORS ALLOWED.
Mathematical Formulae sheet provided

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s) :	D. Nucinkis, D. Nucinkis
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EE1-10A MATHEMATICS I

1. a) Express in the form $x+iy$: [6]

$$(i)\ \frac{\sin\frac{\pi}{3}+i\cos\frac{\pi}{3}}{\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}}, \qquad (ii)\ \left(\frac{i+1}{i-1}\right)^{2019} \qquad (iii)\ (-i)^{-i}$$

b) i) Use the definitions of sinh and cosh in terms of exponentials, to prove that [4]

$$\sinh(x+iy)=\sinh x\cos y+i\cosh x\sin y.$$

ii) Hence, or otherwise, find all complex roots of [5]

$$\sinh^2(z)=-1.$$

c) Find the following limits [6]

$$(i)\ \lim_{x\to 3}\frac{\sqrt{x-2}-1}{\sqrt{x^2-5}-2}, \qquad (ii)\ \lim_{x\to\infty}\left(\frac{x+1}{x}\right)^x, \qquad (iii)\ \lim_{x\to 0}\frac{\ln(1-2x^2)}{e^x-x-1}$$

d) A function is defined as

$$h(x)=\begin{cases} x^{-3}\tan^3(2x) & \text{for } x\neq 0; \\ m & \text{for } x=0, \end{cases}$$

What value of m makes $h(x)$ a continuous function? [4]

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2. a) Differentiate to obtain $\frac{dy}{dx}$: [6]

(*i*) $y=\frac{\ln x}{xe^x}$, (*ii*) $y=\ln(x+\sqrt{x^2-1})$, (*iii*) $x^2+y^2+e^{xy}=1$.

b) Using the definition of the derivative as a limit (differentiation from first principles), differentiate $y=\frac{1}{x}$. [4]

c) Given the function $y=e^{x^2/3}$, [9]

(i) Show that $\frac{dy}{dx}=\frac{2}{3}xy$.

(ii) Differentiate the last equation n times, using Leibnitz' formula, and obtain a recurrence relation giving $y^{(n+1)}(0)$ in terms of $y^{(n-1)}(0)$.
[Note: $y^{(k)}(a)$ means y differentiated k times, and evaluated at $x=a$.]

(iii) Hence, or otherwise, evaluate $y^{(5)}(0)$.

d) Prove by induction that for $n \geq 0$,

$$D^n[\sin(ax)] = a^n \sin\left(ax+\frac{n\pi}{2}\right),$$

where D is the differential operator, $D=\frac{d}{dx}$. [6]

3. a) Evaluate the following integrals: [10]

(i) $\int \frac{\sin x-1}{\sin x+1}\,dx$, (ii) $\int \frac{x^2+2}{x^3+x^2-2x}\,dx$, (iii) $\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx$,

b) Integrate to show that [5]

$$\int \sqrt{1-x^2}\,dx = \frac{1}{2}x\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}x+C.$$

c) Given the function $f(x)=\sqrt{1+x}$,

(i) Obtain the first four terms of the Maclaurin series of $f(x)$, and the Lagrange form of the remainder term R_3; [5]

(ii) If we use the series obtained in (i) to estimate $\sqrt{1.01}$, what is the maximal error incurred? Explain your reasoning. [5]

4. a) Find the radius of convergence of the following series: [7]

$$(i)\quad \sum_{n=1}^{\infty} \frac{\ln n}{n+2} x^n \qquad (ii)\quad \sum_{n=1}^{\infty} \frac{x^{3n}}{n!}$$

b) Given the function defined on the interval $[0,1)$ as

$$f(t) = \begin{cases} t & 0 \le t < 1/2 \\ 0 & 1/2 \le t < 1 \end{cases}$$

i) If $g(t)$ is the *odd* extension of $f(t)$, sketch $g(t)$ for $-2 \le t \le 2$. [2]

ii) Show that the first eight terms of the Fourier Series for $g(t)$ are as follows [9]

$$g(t) = \frac{2}{\pi^2}\sin(\pi t) + \frac{1}{2\pi}\sin(2\pi t) - \frac{2}{(3\pi)^2}\sin(3\pi t) - \frac{1}{4\pi}\sin(4\pi t)$$

$$+\frac{2}{(5\pi)^2}\sin(5\pi t) + \frac{1}{6\pi}\sin(6\pi t) - \frac{2}{(7\pi)^2}\sin(7\pi t) - \frac{1}{8\pi}\sin(8\pi t) + \cdots$$

iii) Choosing an appropriate value for t in the Fourier series for $g(t)$ above, obtain the value of the infinite sum [7]

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} \cdots$$

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